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Keiko Abe
Panasonic Holdings Corporation
Takayuki Kurata
Panasonic Holdings Corporation
Yasuhito Sato
Panasonic Holdings Corporation
Yoshihiko Namba
Panasonic Holdings Corporation
Keisuke Kokubun
kokubun.keisuke.6x@kyoto-u.jp

Kyoto University
Kiyotaka Nemoto
University of Tsukuba
Maya Okamoto
Kyoto University
Yoshinori Yamakawa
Kyoto University

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Development of digital technology to estimate standardized brain health from facial expressions and its application to neuroaesthetics research

Keiko Abe [1], Takayuki Kurata [1], Yasuhito Sato [1], Yoshihiko Namba [1*], Keisuke Kokubun [2*], Kiyotaka Nemoto [3], Maya Okamoto [2], and Yoshinori Yamakawa [2,4,5,6,7]

[1] Product Analysis Center, Panasonic Holdings Corporation, Osaka, 571-8686, Japan
[2] Graduate School of Management, Kyoto University, Kyoto 606-8501, Kyoto, Japan
[3] Department of Medical Informatics and Management and Psychiatry, Institute of Medicine, University of Tsukuba, Tsukuba 305-8575, Ibaraki, Japan
[4] Institute of Innovative Research, Institute of Science Tokyo, Meguro 152-8550, Tokyo, Japan
[5] ImPACT Program of Council for Science, Technology and Innovation, Cabinet Office, Government of Japan, Chiyoda 100-8914, Tokyo, Japan
[6] Office for Academic and Industrial Innovation, Kobe University, Kobe 657-8501, Hyogo, Japan
[7] Brain Impact, Kyoto 606-8501, Japan
* Correspondence: kokubun.keisuke.6x@kyoto-u.jp

Abstract: In recent years, the effects of reading and art on the human brain have been attracting attention. However, it has not been clarified how they affect the brain structure of healthy middle-aged people. This is due to the lack of technology to easily measure the state of brain structure. Therefore, in Study I, by analyzing the correspondence between gray matter volume (GMV) calculated from MRI images of the brain and facial expression data when four types of facial expressions are made (happiness, anger, sadness, and surprise), we constructed an algorithm to estimate brain health from facial expressions and developed a digital application to estimate brain health. In study II, we conducted correlation analysis between the estimated GMV and creative activities and reading habits measured by questionnaire using 113 middle-aged men and women who considered themselves healthy. It was revealed that GMV of the default mode network (DMN) and central executive network (CEN) correlate with creative activity and reading habit, respectively. This is the first study to show that art and reading can affect the brain structure of more people than previously believed using the estimated GMV.

Keywords: art; estimated gray matter volume; facial expressions, reading; triple network.

1. Study I: Development of digital technology to estimate standardized brain health from facial expressions

1.1. Introduction

In recent years, in developed countries, the reduction of medical and nursing care costs due to the decline in the working-age population and the aging of society has become a social issue, and extending healthy life expectancy is considered to be an important factor. In addition, lifestyle habits, aging, and mental and neurological diseases account for a large portion of medical expenses, and there is a demand for the establishment of health care services that improve lifestyle habits. In the healthcare field, inferring a patient's health through face-to-face meetings is common, but it is not necessarily scientific.

A study by Cannavacciuolo et al. [1] compared facial emotional expressivity in patients with Parkinson's disease, Alzheimer's disease, and healthy controls, and found that patients with Parkin-son's disease and Alzheimer's disease shared a common tendency to have lower facial emotional expressivity compared to controls [1]. These symptoms are associated with atrophy of cortical and subcortical areas in the limbic system [2]. Inability to regulate emotional facial expressions can have a negative impact on an individual's socio-emotional environment and has been linked to depression, schizophrenia [3], flattened affect [4], and reduced levels of trait empathy in healthy adults [5]. Alterations in socioemotional attitudes, including deficits in spontaneous emotion expression, empathy, emotion reading, and expression of blunt emotion, are also found in patients with behavioral variant frontotemporal dementia (bvFTD), right temporal frontotemporal dementia (rtvFTD), semantic variant primary progressive aphasia (svPPA), and Alzheimer's disease (AD) [6,7,8,9,10,11]. Impaired intentional emotion expression was predicted by general expressiveness and emotion reading deficits across neurodegenerative patients, and imitation abilities were significantly impaired in bvFTD and rtvFTD patients compared to controls [12].

Blunted facial affect, or reduced facial ex-pression of emotion, is a transdiagnostic com-ponent of serious mental illness [13,14,15,16,17]. Blunted facial affect is a key component of blunted affect, which involves reduced facial expression through many channels, including facial, vocal, and gestural expressions. Blunted affect is a symptom of schizophrenia and is also commonly seen in mood disorders (e.g., psycho-motor retardation in depression), autism spectrum, and neurodegenerative disorders [17,18]. Blunted facial affect is associated with poorer social and occupational functioning [19,20,21] and increased risk of suicide [22]. Therefore, in recent years, research has been conducted using AI technology to identify facial expression features characteristic of people with severe mental illness [23], identify depression using facial expressions and images [24,25], and predict neural networks from facial expressions [26].

On the other hand, there is a certain accumulation of research on predicting brain age from GM images [27]. The difference between brain age and chronological age is quantitatively associated with several age-related risk factors and general physical health, such as reduced grip strength, reduced lung function, history of stroke, increased frequency of alcohol consumption, and increased risk of death [28], as well as declines in cognitive functions such as fluid intelligence, processing speed, semantic language fluency, visual attention, and cognitive flexibility [29,30,31]. Brain age prediction models tend to perform better using GMV than white matter volume [32]. Therefore, it is possible to predict the relationship between facial expressions, which have been shown to be related to dementia and mental illness, and brain age, as expressed by GMV. However, no algorithms have been developed to predict GMV from facial expressions. Considering the cost and time required to measure brain state using MRI, it is difficult to measure frequently, and the development of technology to easily measure GMV from facial expressions could greatly contribute to people's health and welfare.

To date, the authors have been developing evaluation technologies to visualize and quantify the state of the human mind and body, including human senses, emotions, and physical strain, with the mission of contributing to society by creating better products and services through the use of analytical technology to add new value to our customers' products and services. They also possess original facial expression analysis technology [33]. On the other hand, focusing on the "brain" that controls the human mind and body, Yamakawa et al. [34] have proposed the Brain Healthcare Quotient (BHQ) as an index to measure brain health [35]. The BHQ was internationally standardized in 2018 [36] and is easy to use even for non-experts, so it is expected to be used in a variety of solutions, such as allowing users to understand their own brain health and take action to improve their

lifestyle habits. However, the BHQ is an index calculated from Magnetic Resonance Imaging (MRI) images of the brain, and requires MRI imaging at a hospital or other facility, so users cannot easily measure it.

To solve this problem, we used our facial expression analysis technology to analyze the correspondence be-tween BHQ and facial expression data, and discovered the relationship between the two. Using the results, we built an algorithm to estimate BHQ from facial expression data, and developed a digital application that allows users to easily and quickly learn about their standardized brain health.

1.2. Materials and Methods

Data was acquired to analyze the relationship between the BHQ calculated from brain MRI images and facial expression measurement data.

1.2.1. Participants

A total of 58 healthy male and female subjects (47 males and 11 females) aged 25 to 63 years were randomly se-lected from employees of Panasonic Holdings Corporation. Data acquisition was performed after obtaining informed consent from all subjects. According to self-report, none of the recruited participants had a record of neurological, psychiatric, or other medical conditions that could affect the central nervous system. This study was approved by the Ethics Committee of the Institute of Science Tokyo (approval numbers 2023137), and all methods were carried out in accordance with the relevant guidelines, regulations, and principles of the Declaration of Helsinki. All participants provided written informed consent before participation, and anonymity was main-tained.

1.2.2. BHQ indicators

The BHQ mentioned above was used as an index of brain health. There are two types of BHQ cal-culated from brain MRI images: GM-BHQ (Grey Matter-BHQ), which indexes the amount of gray matter where many brain neurons are present, and FA-BHQ (Fractional Anisotropy-BHQ), which in-dexes the quality of white matter where many brain nerve fibers are present [37]. GM-BHQ represents the average of 116 regions in the AAL atlas [38]. We also calculated the average values for regions associated with the Default Mode Network (DMN), Central Executive Network (CEN), and Salience Network (SN). The AAL atlas regions used for each network are shown in Table 1.

Table 1. Network and region Name.

Network	AAL Code	Region Name
DMN	AAL023	Superior Medial Frontal Gyrus (Left)
	AAL024	Superior Medial Frontal Gyrus (Right)
	AAL035	Posterior Cingulate Gyrus (Left)
	AAL036	Posterior Cingulate Gyrus (Right)
	AAL061	Inferior Parietal Lobule (Left)
	AAL062	Inferior Parietal Lobule (Right)
	AAL067	Precuneus (Left)
	AAL068	Precuneus (Right)
CEN	AAL003	Superior Frontal Gyrus (Left)

	AAL004	Superior Frontal Gyrus (Right)
	AAL059	Superior Parietal Lobule (Left)
	AAL060	Superior Parietal Lobule (Right)
SN	AAL029	Insula (Left)
	AAL030	Insula (Right)
	AAL031	Anterior Cingulate Gyrus (Left)
	AAL032	Anterior Cingulate Gyrus (Right)

1.2.3. MRI Data Acquisition

A 3 Tesla MRI scanner (MAGNETOM Prisma, Siemens, Munich, Germany) with a 32-channel head array coil, three-dimensional (3D) T1-weighted magnetization-prepared rapid acquisition gradient echo pulse sequence, and spin-echo echo-planar imaging (SE-EPI) with generalized autocalibrated partially parallel acquisition (GRAPPA) were used for MRI data collection and structural imaging. The following parameters were used: repetition time (TR) 1900 ms, echo time (TE) 2.52 ms, inversion time (TI) 900 ms, flip angle 9°, matrix size 256 × 256, field of view (FOV) 256 mm, and slice thickness 1 mm.

1.2.4. MRI Data Analysis

T1-weighted images were segmented into gray matter (GM), white matter (WM), and cerebrospinal fluid (CSF) using Statistical Parametric Mapping 12 (SPM12; Wellcome Trust Center for Neuroimaging, London, UK) running on MATLAB R2022b (MathWorks Inc., Sherborne, MA, USA). Segmented GM images were then spatially normal-ized using diffeomorphic anatomical registration using the exponential lie algebra (DARTEL) algorithm [39], which included incorporating a modulation step in-to the preprocessing model and smoothing images using a Gaussian kernel with a full width at half maximum (FWHM) of 8 mm. The smoothed GM images were then converted to proportional GM images by dividing by the intracranial volume (ICV) and used to create mean and standard deviation (SD) images. By averaging these in-formation and the local GM quotients using the automatic anatomical labeling (AAL) atlas [40], the GM-BHQ was created with a mean of 100 and a SD of 15 points.

Previous research has confirmed that BHQ tends to decline with age and can be an indicator of the youthfulness of the brain6). Research into GM-BHQ's relationship with lifestyle has also been conducted; for example, resting and doing housework on weekdays has a positive effect on GM-BHQ, while taking too much time off on weekends tends to be bad for GM-BHQ. It has been shown that lifestyle is related to brain health, and it is hoped that reviewing lifestyle habits will contribute to maintaining and improving brain health6). In addition, in previous studies, the whole-brain GM-BHQ has been related to dietary balance [41], lifestyle [42], social performance [43], curiosity and grit [44], and diversity understanding [45]. These studies were conducted in healthy middle-aged adults, not elderly people or people with underlying diseases, so it is expected that using the same index will add depth to the interpretation of the study. Our study, which focuses on GMV, a function of surface area and cortical thickness, is more comprehensive than studies using other neural biomarkers such as functional MRI (fMRI), which is excel-lent for examining brain responses [46], and cortical thickness, which is suitable for observing genetic influences [47].

1.3. Analysis

Two types of measurements were performed: MRI brain imaging and facial expression measurement.

(a) MRI brain imaging

With the cooperation of Brain Impact, a general incorporated association, brain MRI imaging was performed twice, in July and December 2023.

(b) Facial expression measurement

For subjects who underwent brain MRI imaging, facial expression measurements were also performed at the same time, with 37 subjects being measured twice, in July and December, and 21 subjects being measured once, in July or December.

For facial expression measurement, an original PC ap-plication developed by the authors was used. This application imitates facial expressions in photos displayed on a PC screen in front of a camera, and as shown in Figure 1, there are four types of facial expressions: joy, anger, sadness, and surprise. The photos were displayed randomly for each subject, and facial expression data was obtained from the subjects when they continued to imitate each facial expression for 5 seconds.

Figure 1. Facial expression measurement application.

A total of 95 sets of data from the same subject, each consisting of BHQ from MRI brain images and facial expression data, were analyzed.

(a) BHQ from MRI brain images

With the cooperation of the Brain Impact Association, GM-BHQ values for the entire brain and for each brain region, DMN, CEN, and SN, were calculated from MRI brain images and used for analysis.

(b) Facial expression measurement data

468 facial landmarks were detected from the measured facial expression data (see Figure 2), and the time when the landmark change was maximum for each facial expression when imitating four types of facial expressions, joy, anger, sadness, and surprise, was used as the reaction time for analysis.

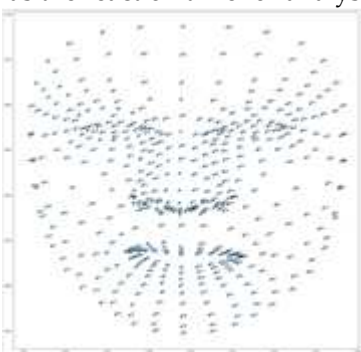

Figure 2. Facial landmarks.

1.4. Results

1.4.1. Confirmation of the relationship between BHQ and age

As mentioned above, previous research has confirmed that BHQ tends to decrease with age. First, we used BHQ data from brain MRI images of 362 healthy men and women aged 22 to 67 (262 men, 100 women) that we had obtained separately to confirm the relationship between BHQ and age.

As an example, Figure 3 shows the relationship between whole-brain GM-BHQ and age. The correlation coefficient between whole-brain GM-BHQ and age was 0.71, indicating a strong correlation.

In analyzing the relationship between whole-brain GM-BHQ and facial expressions, we therefore decided to eliminate the effect of age, that is, to use the difference of GM-BHQ from the age average (approximation line) (hereinafter referred to as ΔGM-BHQ) as the objective variable and to search for the facial expression parameters that are optimal for explaining ΔGM-BHQ.

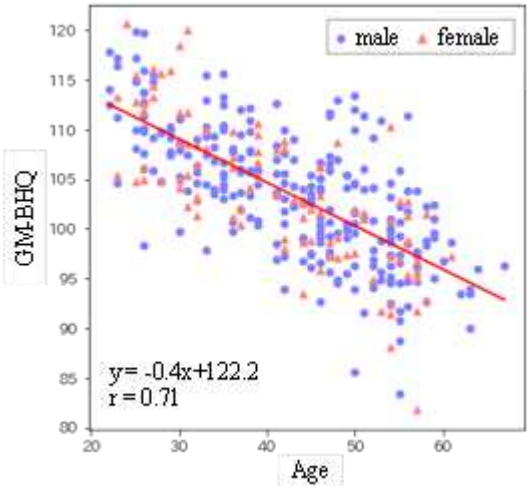

Figure 3. Relationship between GM-BHQ and Age.

We also checked the relationship between age and GM-BHQ in the DMN, CEN, and SN brain regions, finding that the correlation coefficients between the two were 0.67, 0.61, and 0.67, respectively, confirming that there is a correlation between age and GM-BHQ in the entire brain.

Based on these results, we used a similar approach to extract facial expression features that estimate each Δ component for the DMN, CEN, and SN, using ΔDMN, ΔCEN, and ΔSN as objective variables.

1.4.2. Building a BHQ estimation model

Using each Δ component as the objective variable and the reaction time in the X and Y directions of each facial landmark as the variables, the analysis was carried out in the following steps to extract facial expression features that can estimate each Δ component.

Step 1: Extract variables correlated with the objective variable.

Step 2: In order to exclude variables with multicollinearity, variables with high correlation coefficients were grouped together, and only the variable most correlated with the objective variable for each group was extracted.

Step 3: Construct a linear multiple regression model using the forward variance method.

In selecting the model, cross-validation was performed on linear regression models, linear mixed models, and random forest models using the Leave One Group Out

method, and it was decided to adopt the linear regression model with the highest accuracy.

1.4.3. Extracted features

As a result of the analysis, Figure. 4 shows an example of facial expression features extracted to estimate ΔGM-BHQ. Figure 4 shows the facial landmarks that are key when imitating the expression of "happiness," and it can be seen that the area around the eyes is important.

Figure 5 also shows the percentage of people who showed a strong correlation with the happiness scale, with a correlation coefficient of 0.7 or more, when imitating the expression of "happiness," with darker colors indicating a higher percentage showing a stronger correlation. We can see that when imitating the expression of "happiness," many people make the expression with movements around the mouth, and the percentage of people who make the expression with movements around the eyes tends to be lower. From the above, it can be assumed that people who are able to imitate facial expressions while moving parts of the face that are difficult for most people to move as widely and quickly as possible, rather than parts of the face that are easy to move, have a high ΔGM-BHQ.

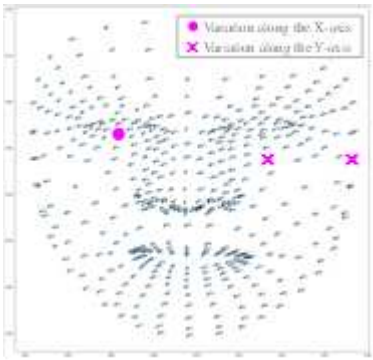

Figure 4. Key facial landmarks when expressing happy.

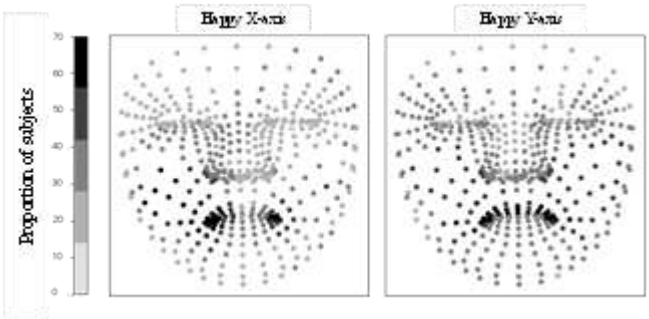

Figure 5. Proportion of subjects showing strong correlation with scale of happy.

As a result of the analysis, five items were extracted for ΔGM-BHQ, four items for ΔDMN, eight items for ΔCEN, and six items for ΔSN as facial expression features for estimating each Δ component, and a linear multiple regression model was constructed.

1.4.4. Accuracy of whole brain GMV estimation

Figure 6 shows the results of estimating the GM-BHQ of the entire brain using an estimation model constructed using the extracted facial expression features. The correlation coefficient between the GM-BHQ from brain MRI images and the GM-BHQ

estimated from facial ex-pressions was 0.75, confirming a strong correlation. In addition, the mean absolute error (MAE) for the GM-BHQ from brain MRI images was 3.12. For the 95 data sets analyzed in this study, the correlation coefficient between the value when the GM-BHQ of the entire brain was estimated from age alone and the GM-BHQ from brain MRI images was 0.66, and the MAE was 3.54, so it is believed that adding facial ex-pression features contributes to improving the estimation accuracy.

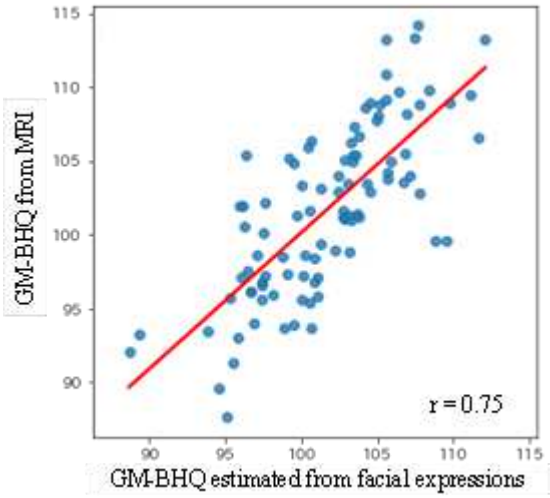

Figure 6. Relationship between GM-BHQ from MRI and GM-BHQ estimated from facial expressions.

1.4.5. Accuracy of DMN, CEN, and SN estimation

Similarly, the results of estimating the GM-BHQ of each brain region, DMN, CEN, and SN, are shown in Figuers. 7 to 9. The correlation coefficients between the DMN, CEN, and SN from brain MRI images and those estimated from facial expressions were 0.58, 0.63, and 0.74, respectively, confirming the existence of a correlation. In addition, the MAE for the DMN, CEN, and SN from brain MRI images was 4.14, 3.87, and 4.05, respectively. For the 95 data analyzed in this study, the correlation coefficients between the values of DMN, CEN, and SN estimated from age alone and those from brain MRI im-ages were 0.53, 0.50, and 0.68, respectively, and the MAE was 4.34, 4.25, and 4.53, respectively. Compared to estimation based on age alone, the correlation coefficients were higher and the MAE was also improved, so it is thought that adding facial expres-sion features contributes to improving the estimation accuracy, as with the GM-BHQ of the entire brain. The estimated accuracy results for GM-BHQ of the whole brain and the DMN, CEN, and SN regions are summarized in Table 2.

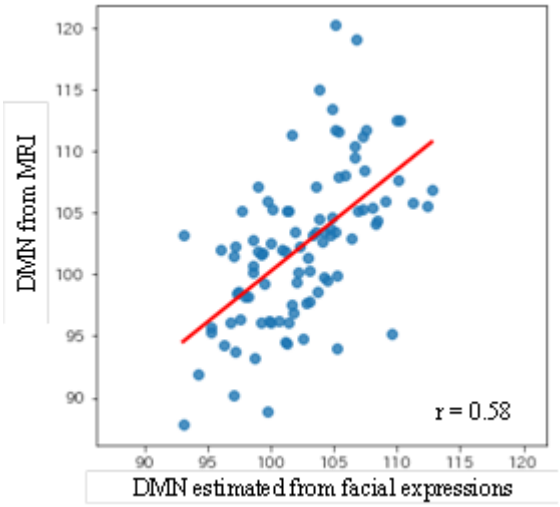

Figure 7. Relationship between DMN from MRI and DMN estimated from facial expressions.

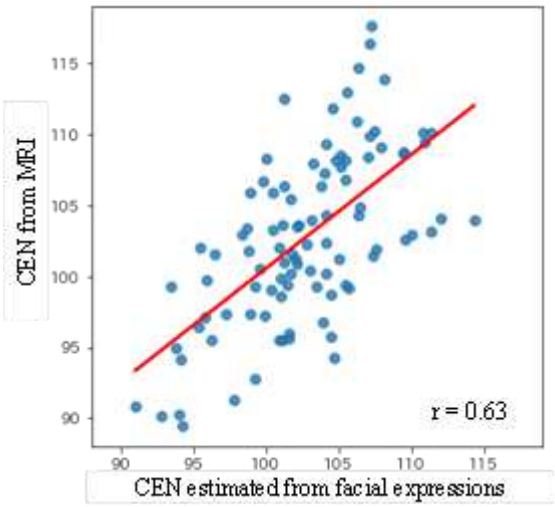

Figure 8. Relationship between CEN from MRI and CEN estimated from facial expressions.

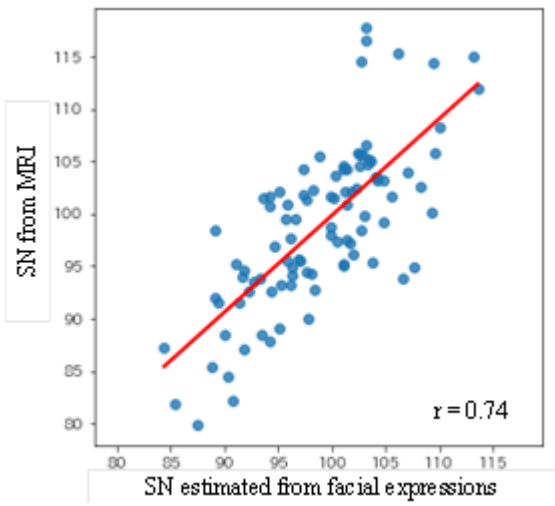

Figure 9. Relationship between SN from MRI and SN estimated from facial expressions.

Table 2. Comparison of estimation results from facial expressions and age.

Regions of brain	Estimated results from facial expressions		Estimated results from age	
	Correlation coefficient	MAE	Correlation coefficient	MAE
GM-BHQ	0.75	3.12	0.66	3.54
DMN	0.58	4.14	0.53	4.34
CEN	0.63	3.87	0.50	4.25
SN	0.74	4.05	0.68	4.53

1.4.6. Developing Applications

Using the estimation model described above, we developed a PC application called "Estimated BHQ" that estimates brain health from facial expressions. With this application, users simply follow the instructions on the screen in front of the camera and imitate four types of facial expressions - happiness, anger, sad-ness, and surprise (see Figure 10) - to easily and enjoyably find out the results of the GM-BHQ, which measures the health of the entire brain, as well as the GM-BHQ results for the DMN, CEN, and SN regions (see Figure 11). This allows users to easily and quickly measure their brain health and brain characteristics.

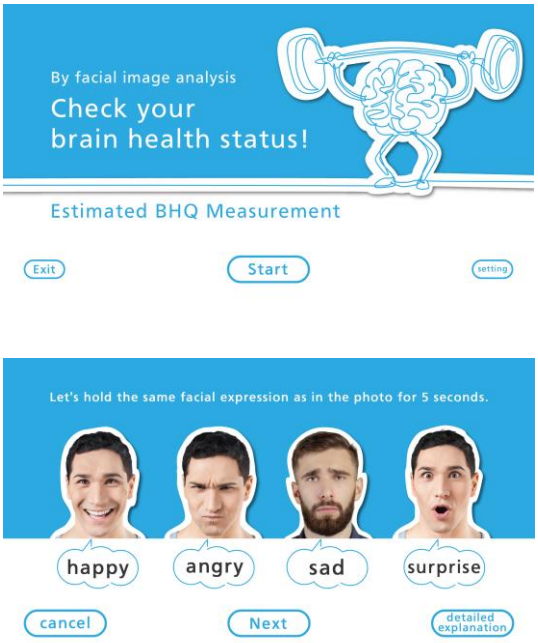

Figure 10. “Estimated BHQ” application.

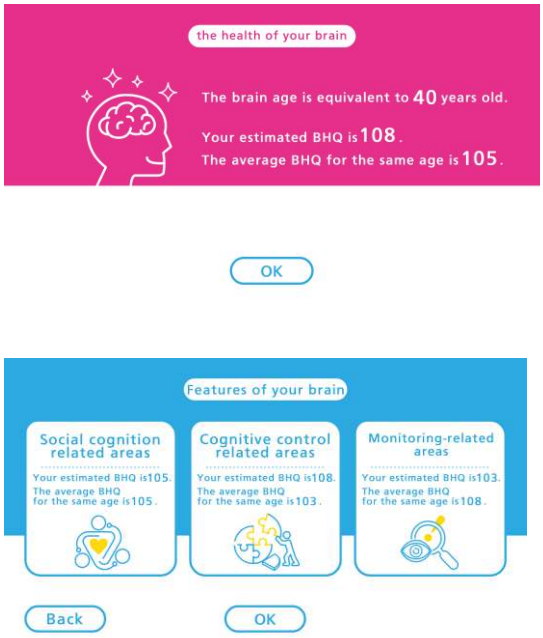

Figure 11. Measurement results.

1.5. Discussion

By analyzing the correspondence between GM-BHQ, which indicates the state of brain health, and facial ex-pression data when four types of facial expressions (happiness, anger, sadness, and surprise) were made, we were able to extract facial expression features that are important for estimating GM-BHQ. In addition, we constructed an estimation model using these features to estimate the GM-BHQ of the entire brain, and the correlation coefficient between the GM-BHQ from brain MRI images and the GM-BHQ estimated from facial expressions was 0.75, and the MAE was 3.12. Similarly, when we estimated the GM-BHQ of each brain region, the DMN, CEN, and SN, the correlation coefficients between the DMN, CEN, and SN from brain MRI images and the DMN, CEN, and SN estimated from facial expressions were 0.58, 0.63, and 0.74, respectively, and the MAE was 4.14, 3.87, and 4.05, respectively, although the estimation accuracy was lower than that of the GM-BHQ of the whole brain.

Furthermore, we developed a PC application called "Estimated BHQ" that estimates brain health from facial expressions, and have conducted various demonstration experiments to confirm user needs and acceptability for measuring brain health [48,49,50]. Through these demonstration experiments, we have demonstrated the possibility that by providing unprecedented new value of being able to measure brain health in a fun and easy way, we can arouse users' interest in brain health, and by visualizing brain health, we can encourage them to try measuring it again. These results suggest that measuring brain health can be a catalyst for encouraging behavioral changes in users.

Currently, it is possible to visualize brain health, but it is not yet possible to present specific actions on how users can maintain and improve their brain health based on the measurement results. In the future, we plan to explore measures to improve brain health while also pro-posing improvement actions tailored to each user. This research is the first to develop technology to predict whole-brain and triple network GMV from facial expressions. We hope to continue developing brain health estimation technology and proposing measures to improve brain health, thereby contributing to maintaining and improving people's brain health and realizing well-being in their lives.

1.6. Limitations

The current research has some limitations. The algorithm developed in this study is intended to obtain an estimate of GMV, not an exact value. Therefore, it should be used with caution in research that requires greater accuracy, such as in the medical field. In addition, this study was conducted on Japanese people, so the results cannot be applied directly to other countries. At the same time, as symbolized by the bias in the gender ratio of the sample, the results of this study do not reflect the average image of Japanese people. In the future, the results of this study should be verified using a sample with a larger number of diverse backgrounds.

2. Study II: An application of estimated GMV to neuroaesthetics research

2.1. Introduction

In recent neuroscience, the effects of artistic activities and reading have been attracting attention as a way to reduce the risk of dementia in the elderly. Sensing beauty activates the entire brain, including the brain's reward system, such as the frontal orbital cortex and striatum [51,52]. Therefore, it has been reported that there is a relationship between high creativity and several brain regions. In particular, it has been revealed that DMN cooperates with CEN to improve creativity and artistic performance [53,54]. Creative activities may have a positive effect on the brain not only by performing them yourself, but also by watching and feeling the state of the creative activity. It has been reported that the DMN is engaged when people are deeply moved by a work of art or when they feel it is beautiful [55,56]. On the other hand, when learning something from literature, we connect various events and imagine the mental scenes of people at that time. Such cognitive acts are related to the activity and connectivity of the hippocampus and prefrontal cortex [57,58]. Previous evidence suggests that there is a relationship between the retrieval of autobiographical memories and CEN [59,60]. However, the effects of such artistic activities and reading on the brain structure of healthy middle-aged adults remain unclear. The possibility that engaging in intellectual activities at a young age may prevent or delay the onset of dementia has been asserted and supported by research on "cognitive reserve" [61,62,63].

Therefore, in this study, we focus on the GMV of the Healthy adult brain. GMV was determined as the GM-BHQ value originally based on MRI image analysis. This value can be estimated by measuring facial expressions as shown in Study I. The representative brain networks are three networks that control rational judgment, emotion control, other recognition, self-recognition, and behavior evaluation. Of these, CEN, composed of the dorsolateral prefrontal cortex and posterior parietal cortex, is critical for the active retention and manipulation of information in working memory, attention, problem solving, decision-making, and self-awareness [64,65]. The SN is also a network that includes the ventrolateral prefrontal cortex (VLPFC) and anterior insula (collectively referred to as the fronto-insular cortex FIC), as well as the anterior cingulate cortex (ACC) [66], and responds to subjective salience, whether cognitive, homeostatic, or emotional [67]. The SN also acts as a switch between the CEN and the DMN, inhibiting the latter and activating the former when salient stimuli or cognitive tasks are at hand, a process essential for attention and flexible cognitive control [68,69,70,71,72,73]. On the other hand, the DMN includes the medial posterior cortex, including the posterior cingulate cortex (PCC) and parts of the frontal lobe, the medial prefrontal cortex (MPFC), and the posterior temporal region around the temporoparietal junction (TPJ), including the inferior parietal lobe (IPL)

[74,75,76]. The DMN is preferentially activated when individuals are not focused on the external environment [75] and are engaged in various areas of cognitive and social processing. That is, the medial prefrontal cortex (MPFC) plays a key role in the social understanding of others, and the connections between the medial prefrontal cortex (aMPFC) and the posterior cingulate cortex and anterior cingulate cortex mainly contribute to self-other discrimination. On the other hand, the connection between the dorsal MPFC (dMPFC) and the temporoparietal junction (TPJ) is mainly related to the understanding of others' mental states [77].

The interactions between these networks are related to daily self-regulation and empathy for others [78,79,80,81,82,83]. However, to our knowledge, there have been no studies that have addressed the relationship between the brain structure centered on these triple networks and artistic activities and reading habits. Therefore, in this study, we contribute to the development of related research by clarifying the characteristics of the brain structure centered on the triple network in healthy adults using information obtained from questionnaire surveys and estimation of GMV using facial expressions.

2.2. Materials and Methods

2.2.1. Participants

The sample size required to perform correlation analysis at a significance level of 5% and power of 80%, with an effect size of r = 0.3, which corresponds to a "moderate correlation" according to Cohen's criterion, was calculated as 109 samples using G*Power 3.1.9.7. A total of 113 people (58 men, 54 women, and 1 unknown) participated in the study, with a mean age of 39.0 ± 10.1 years. Participants were randomly selected from among employees of a single private company and asked to participate. Participants completed online questionnaires and facial expressions measurement from March to April 2025. According to self-report, none of the participants had a history of neurological, psychiatric, or other medical conditions that could affect the central nervous system. All methods were carried out in accordance with relevant guidelines and regulations, and all participants provided written informed consent before participation, and anonymity was maintained. This study was approved by the Institute of Science Tokyo's Brain Information Cloud (Ethics Committee for Human Research: Permit Number 2023137) Ethics Committee. Table 3 and Table 4 show information on various variables of participants.

Table 3. Participant information (frequency).

	N	%
Sex		
Male	58	51.3
Female	54	47.8
No answer	1	0.9
Position		
Executives/executives	6	5.3
Department head	2	1.8
Manager/leader	40	35.4
General/professional	62	54.9
Contract/individual business	3	2.7
Occupation		

Administrative occupations	17	15
Research/technical occupations	9	8
Professional occupations such as legal affairs, management, culture/arts, etc.	17	15
Administrative occupations	26	23
Sales/sales occupations	5	4.4
Service occupations	39	34.5

Table 4. Participant information (minimum, maximum, and average value).

	Mean	N	SD	Max	Min
Age	39.0	113	10.1	24	68
BMI	22.8	113	4.5	17.2	55.2
Years of education	16.7	113	2.0	6	22
Years of service	3.5	113	1.7	1	6

SD: standard deviation.

2.2.2. Estimating brain information using facial expression information

In this study, we used estimated GM-BHQ measured from facial expressions. To measure facial expressions, we used an application developed in Study I. The reason for using the estimated GM-BHQ is that GM-BHQ has been used primarily in studies of healthy middle-aged adults, as in this study, rather than in elderly people or people with underlying diseases, and correlations with various psychological indices have been confirmed. In our previous studies, whole-brain GM-BHQ was positively correlated with cognitive ability [84], job satisfaction [85], dietary balance [41] and negatively with unhealthy lifestyle [42], fatigue and stress [86], and polluted residential environment [87]. Furthermore, in a recent pilot study, the DMN, CEN and SN of GM-BHQ were positively correlated with understanding of gender and origin diversity [45]. Also, consistent with neuroplasticity claims [88], GM-BHQ has been shown to change with certain interventions [89,90].

2.3. Psychological Test

2.3.1. Creative Behavior

This study used the Aesthetic Responsiveness Assessment developed by Schlotz et al. [91] and adapted by Magsamen and Ross [92]. The AReA originally is a short (14-item) questionnaire for assessing how responsive a person is to aesthetic experiences, such as with visual arts, music, dance and performance. Among the three subscales of AReA, namely Aesthetic Appreciation (8 items), Intense Aesthetic Experience (3 items), and Creative Behavior (3 items including "I sculpt, paint, draw, direct films, or do design work"), we used a revised creative behavior scale consisting of four items: original Creative Behavior three items plus one item from Aesthetic Appreciation, "I visit museums or go to musical/dance performances." This revision was made because the added item is an item that represents behavior, like the three items in Creative Behavior, and a factor analysis using test data from 80 people collected through general recruitment showed that the

factor was more stable when it was included in Creative Behavior rather than Aesthetic Appreciation, and also because an improvement was observed in the Cronbach's alpha reliability coefficient of the factor (available upon request). Each item was answered on a 5-point Likert scale, and each response was assigned a score from 1 to 5. The creative behavior was calculated as the average of 4 items. Cronbach's alpha reliability coefficients of creative behavior were 0.662.

2.3.2. Reading habits

The study used two items from the adult version of the OECD Programme for International Student Assessment (PISA): "How often do you read books in your current job?" and "How often do you read novels or non-fiction books outside of work?" [93]. Each item was answered on a 5-point Likert scale, and each response was assigned a score from 1 to 5. The creative behavior was calculated as the average of 2 items.

2.3.3. Control variables

Sex (male = 1; female = 0), age (years old), educational background (years) and BMI (kg/m^2) were used as control variables.

2.4. Data Analysis

Correlation and partial correlation analyses were performed between the estimated whole-brain and regional GMVs of the CEN, SN, and DMN, creative activities and reading habits variables. The standard for statistical significance was 5% two-sided. All statistical analyses were performed using IBM SPSS Statistics Version 28 (IBM Corp., Armonk, NY, USA).

2.5. Results

The figures below the diagonal shows the Pearson correlation coefficient, and the figures above the diagonal shows the Pearson partial correlation coefficient. The correlation coefficients did not show any significant correlation between GMV and the creative activities and reading habits variables. Therefore, the results of the partial correlation coefficients are shown below. DMN was positively and significantly correlated with creative behavior (r = 0.231, p = 0.016) but was not significantly correlated with reading habits (r = 0.095, p = 0.329). CEN was positively and significantly correlated with reading habits (r = 0.231, p = 0.016) but was not significantly correlated with creative behavior (r = -0.028, p = 0.777). Figure 2 and Figure 3 show the results of these significant correlations in scatter plots. There were no significant correlations between other GMV and creative behavior or reading habits.

Table 3. Correlation coefficient.

		1	2	3	4	5	6
1	BHQ		0.650***	0.568***	0.737***	-0.012	0.186
2	DMN	0.923***		0.643***	0.610***	0.095	0.231*
3	CEN	0.858***	0.878***		0.617***	0.231*	-0.028
4	SN	0.940***	0.910***	0.872***		0.076	0.166
5	Reading habits	-0.074	-0.042	0.060	-0.044		0.253**
6	Creative behavior	0.164	0.187*	0.065	0.167	0.195*	

N = 113; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$. The figures below the diagonal are Pearson's correlation coefficients. The figures above the diagonal are Pearson's correlation coefficients controlling for demographic variables.

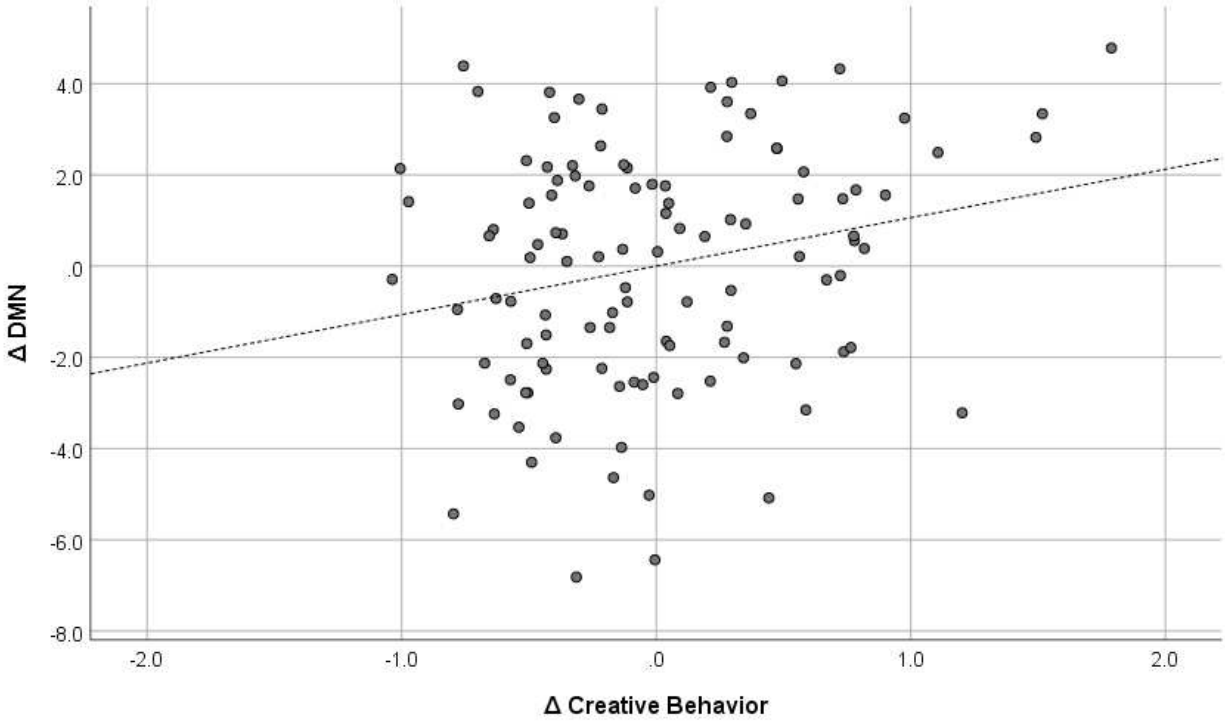

Figure 2. Creative Behavior and DMN (r = 0.231). Δ: Residuals from a regression with the control variables as independent variables.

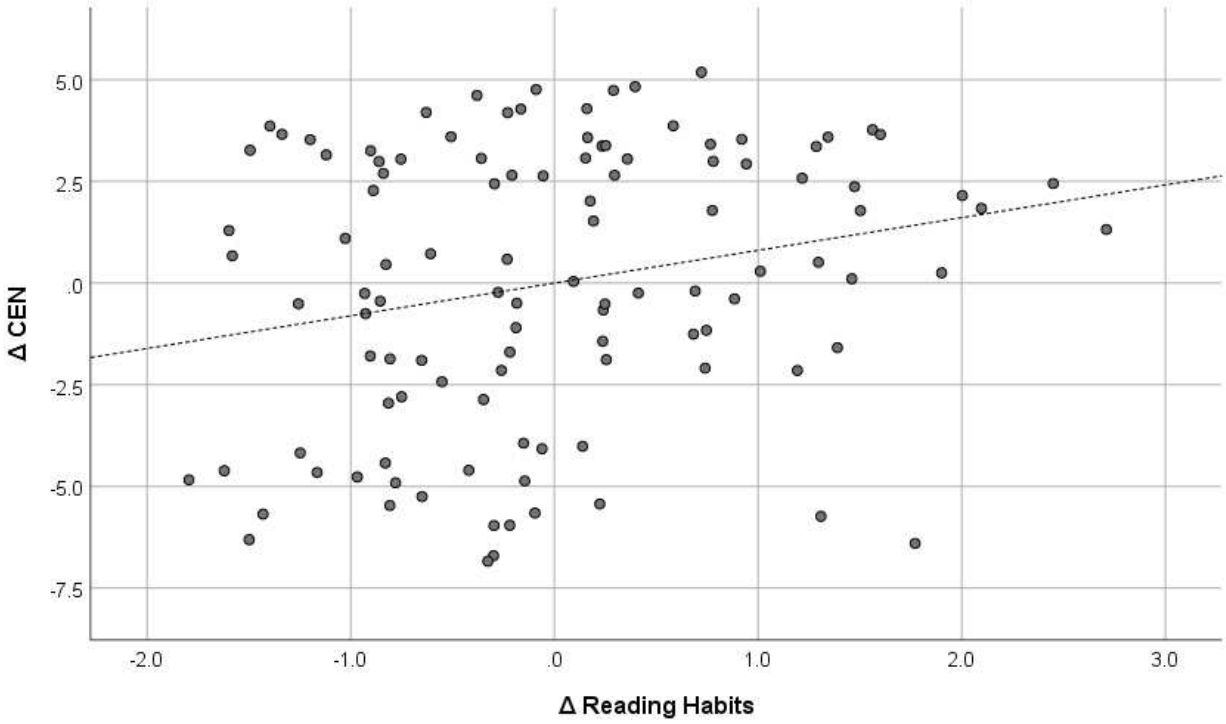

Figure 3. Reading Habits and CEN (r = 0.231). Δ: Residuals from a regression with the control variables as independent variables.

2.6. Discussion

The results of this study suggest that the brain GMV of healthy adults is related to intellectual activities such as creative behavior and reading habits. The DMN and CEN seem to play a major role in these activities. This result is consistent with previous studies that have shown that the DMN is associated with artistic activities [53,54,55,56] and that the CEN is associated with reading [57,58,59,60]. The participants in this study were male and female workers who considered themselves healthy. Therefore, the results of this study indicate that engaging in intellectual activities such as art and reading while young and healthy may not only maintain their brain health and have a positive effect on their daily work, but may also contribute to extending their future healthy lifespan. This idea has been discovered or supported by previous studies as "cognitive reserve" [61,62], but the results of this study add biological knowledge to this discussion.

In a previous study in which 16 young adults were tasked with rating images of artworks, areas of the medial prefrontal cortex, measured in an fMRI scanner, were positively activated in the best-rated trials. This means that the rating-specific neural response is attributable not to the characteristics of the particular artwork, but to the aesthetic experience. In other words, DMN activity is due to the fact that a particular artwork fits the unique structure of the individual so well that it accesses neural substrates that are involved in the self, leading to a feeling of being "moved." This explanation is consistent with the contemporary idea that an individual's taste in art is linked to a sense of identity [94,95]. The involvement of the DMN in painting appreciation is further supported by the results of a recent meta-analysis [96].

On the other hand, an experimental study of the reading comprehension of 60 school-age children showed that executive functions make a significant contribution to reading comprehension [97]. Subsequent meta-analysis also supported a moderate positive association ($r = 0.36$) between executive function and reading comprehension [98]. Consistent with these, a recent neuroscience study of 55 school-age children showed that the discrepancy between the ability to convert printed words into sounds and reading comprehension of text was associated with a frontoparietal network including the GMV of the left dorsolateral prefrontal cortex [99].

The results of this study showing that the DMN is involved in artistic activities and the CEN in reading are highly suggestive. This is because, in light of the accumulation of research showing that whole-brain GMV is involved in cognitive functions [100], it suggests the importance of having a variety of hobbies. Normally, human memories are stored in the hippocampus as short-term memories containing a lot of concrete information. However, because the hippocampus has a capacity, it cannot store the details of concrete information for long. Much of the information gradually loses concreteness and is stored in the brain as abstract information over a long period of time. People will become interested in information they saw on TV yesterday, such as health information, and will search for books driven by the desire to know more about it in more detail. At this time, people will search for books, read, and think while referring to the concrete information in their brains that is still fresh, so they will rely on the executive function of the CEN. However, for information that has been fragmented in the brain and does not rise to the conscious mind, it may be difficult to search for it of their own volition and find relevance in the book they picked up. At that time, people will feel a vague sense of dissatisfaction. However, that person may suddenly start a little artistic activity one day, perhaps inspired by a view they saw casually. In the process of looking at art in a museum or fumbling around to draw a picture by themselves, one day they suddenly come across a piece that moves them in a way that they cannot put into words. At this time, the DMN plays a central role. The DMN responds to aesthetic experiences that they are not aware of and have forgotten. Artistic activities are a little more difficult to get into than books

that can be easily picked up at a bookstore, and many people do not have the opportunity to come into contact with the results of neuroaesthetics research, so at present, it is thought that those who are lucky enough to come across them are involved in art and maintaining their brain health. However, if more people could learn about their brain condition using a machine that can be easily measured, the effect of casually engaging in art on the brain would become clearer, and it may be possible to get more people involved in art and improve their brain health. The results of this study have the potential to contribute to improving the well-being of such people in the future. In particular, reliable information based on abundant evidence is needed to encourage greater participation in artistic activities among elderly people, who are considered to be conservative and unwilling to change their ways of thinking, and to prevent dementia.

Thus, the results of this study are consistent with this series of previous studies, despite the use of a simplified method of facial expression recognition. Previous neuroscience research has required expensive and heavy equipment such as MRI. This has made it difficult for ordinary workers to easily measure their brains. On the other hand, by placing the simple measuring device used in this study in workplaces and commercial facilities and measuring the state of the brain easily and on a daily basis, just as if measuring weight or blood pressure, people may be motivated to pay more attention to their own and others' brain states and maintain brain health. It is also clear from neurofeedback research that knowing the state of the brain leads to maintaining and improving brain health [101,102]. Therefore, the results of this study will make a meaningful contribution to the social implementation of brain science.

2.7. Limitations

This study has some limitations. Firstly, it should be noted that the results of this study were estimated from facial expressions, not GMV calculated from MRI images, so the reproducibility of the results of this study needs to be tested using GMV. Secondly, this study was conducted on Japanese people, and caution is required if the results are applicable to other countries. Japanese people have a unique sense of beauty, such as wabi-sabi and kawaii, and such peculiarities may have biased the results. They are also known to be conservative and, compared to other developed countries, enjoy reading but are not very active in artistic activities. Thirdly, because general measures of art and reading were used, the results of this study cannot suggest specific activities. Fourthly, this study is a cross-sectional analysis and does not show a causal relationship. For example, rather than the arts and reading enhancing brain health, it may be that brain health drives people's choices about the arts and reading. Fifthly, the small sample size limits the generalizability of the results. Sixthly, in this study, sex, age, educational background and BMI were used as control variables. However, background information not adopted in this study, such as socioeconomic status such as income and differences in the community to which the participants belong, may have influenced the results. Beauty is not a distinct, independent element embedded in a color, food, nature, landscape, idea, or even a face, nor is it something an artist "puts" into a work as a distinct entity; it is a property that emerges in the beholder's brain. That aesthetic response is shaped by the beholder's cultural experiences, life events, education, genetic inheritance, and perhaps gender, age, and health. Individual variation in the intensity and extent of the response has not been systematically explored, and there remains much to be explored [103]. Seventhly, this study was unable to clarify the relationship between SN and art and reading. This is an issue that remains for future efforts to improve brain health at the whole-brain level. Eighth, even if art and reading are good for the brain, this study does not allow us to infer the extent to which their effects are compared to other activities. With the advancement of neuroscience, there are many activities that are said to be good for the brain (exercise, diet, brain training, etc.). Since the general public has

limited free time and money, the relative importance of neuroaesthetic-related activities, how often and for how long they are effective, and above what level the incremental effect disappears, need to be gradually clarified in the future. Ninth, the present study was limited to a triple network: a recent meta-analysis showed that neural responses to artistic beauty recruit a broadly distributed network in both hemispheres, including content-dependent brain regions in the ventral visual stream [104]. Previous findings have also shown that the originality of a art is associated with the DMN, while beauty is associated with a network of areas including the occipital and frontal lobes [105]. As such, the latest findings in brain research indicate the possibility of a more diverse and complex network of neural responses to beauty, leaving room for further development of the simplified measuring device used in this study. Finally, most previous studies have used fMRI, which differs from the data of this study, which is based on MRI. Although fMRI is excellent for measuring brain responses, there is insufficient evidence as to whether the responses have a lasting effect on brain structure. As neuroaesthetics research using MRI becomes more active and the relationship with brain structure becomes clearer, the value of the simple measurement used in this study is expected to increase even further. Future research should seek to verify the results of this study using other methods, such as MRI and fMRI, in a longitudinal manner that includes a variety of intervention methods and control variables, with a larger number of people in different cultures, in order to obtain further new findings.

3. Conclusions

In Study I, by analyzing the correspondence between the BHQ calculated from MRI images of the brain and the facial expression data when four types of facial expressions (happiness, anger, sadness, and surprise) are made, an algorithm was constructed to estimate brain health from facial expressions, and an application was developed to estimate brain health. This allows users to easily find out the state of their brain health in a short time by simply measuring their facial expressions.

In Study II, a survey was conducted on 113 healthy Japanese adults using GMV estimated from facial expressions and questionnaires, and it was shown that there was a positive correlation between local DMN and CEN GMV and artistic activities and reading habits, respectively. These results suggest that estimated GMV and art and reading habits are effective in paying attention to, caring for, and intervening in the brains of healthy people to maintain future brain health and reduce the risk of dementia.

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Institutional Review Board Statement: The study was conducted in accordance with the Declaration of Helsinki and approved by the Institute of Science Tokyo's ethical committee (Approval Numbers 2023137 approval on 29 October 2021).

Informed Consent Statement: Informed consent was obtained from all subjects involved in the study.

Data Availability Statement: The data presented in this study are available on request from the corresponding author due to the need to protect the privacy of participants.

Conflicts of Interest: Authors Keiko Abe, Takayuki Kurata, Yasuhito Sato, and Yoshihiko Namba were employed by the company Panasonic Corporation. Author Yoshinori Yamakawa was employed by the company Brain Impact. The remaining authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Abbreviations

The following abbreviations are used in this manuscript:

GM-BHQ	Gray Matter Brain Healthcare Quotient
DMN	Default Mode Network
CEN	Central Executive Network
SN	Salience Network

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